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From opportunity to advantage: how firms can leverage generative AI affordances to drive success in competitive markets

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ABSTRACT

Although firms with a proactive technology stance are increasingly adopting generative artificial intelligence (GenAI) to enhance competitiveness, integrating GenAI into business operations presents significant challenges. To address this dilemma, we ground in affordance theory to explore how firms can leverage GenAI affordances to transform technological opportunism (TO) into sustained competitive advantages, particularly in highly competitive environments. We draw on survey data from 223 firms spanning different industries in China and find that three GenAI affordances—organizational memory, collaborative, and creational affordances—mediate the relationship between TO and competitive advantages. Moreover, competitive intensity amplifies the mediation effect of creational affordance instead of the mediation effects of organizational memory and collaborative affordances. We add to the extant AI-driven business literature by emphasizing the mechanisms behind the TO–competitive advantages link from the perspective of GenAI affordances. Lastly, our findings provide practical guidance for managers aiming to harness GenAI effectively.

1. Introduction

In today's rapidly evolving, technology-driven business environment, adopting a proactive stance toward identifying and leveraging emerging technologies is increasingly critical (Bendig et al., 2023). This proactive stance, commonly referred to as technological opportunism (TO), is not merely about awareness of new technologies but entails a deliberate, forward-looking approach to seeking and applying emerging technological advancements (Lucia-Palacios et al., 2016). Theoretically, firms characterized by TO are well-positioned to seize new opportunities and adapt to evolving market conditions (Bullini Orlandi et al., 2020). However, the transition from TO to achieving sustained competitive advantage is fraught with challenges and complexities (Chen and Lien, 2013). A recent report indicates that 30 % of technology-driven projects are abandoned after the proof-of-concept stage (Gartner, 2024). This finding suggests that merely recognizing TO does not ensure success; firms must effectively deploy and integrate these technologies into their strategic frameworks.

As generative artificial intelligence (GenAI) emerges as a transformative force (Baabdullah, 2024), many firms with high TO actively adopt this technology (Mariani and Dwivedi, 2024). However, these firms also face significant challenges. For example, although GenAI has the potential to revolutionize product development, marketing, and customer interaction (Kshetri et al., 2024), integrating it into existing technological infrastructures often demands substantial investment and process reconfiguration (Fosso Wamba et al., 2023). Furthermore, although GenAI provides valuable insights for decision-making and strategy (Chowdhury et al., 2024), the key challenge lies not only in generating these insights but also in translating them into actionable strategies (Cook et al., 2024). Therefore, to fully capitalize on technological opportunities and sustain competitive advantages, it is essential to understand how firms can embed GenAI into their business processes.

We draw on affordance theory to examine how firms characterized by TO leverage GenAI (Sesay et al., 2024). Specifically, affordance refers to the perceived and actual properties of an object or environment that shape how it can be used or interacted with (De Luca et al., 2021).

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According to prior literature, technology affordances generally include organizational memory affordance (i.e., using technologies to enhance organizational knowledge) and collaborative affordance (i.e., using technologies to foster cooperation within organizations (Chatterjee et al., 2020). Because GenAI shares characteristics with general technologies, such as enhancing the ability to store, retrieve, and apply knowledge through data processing and analysis (Sullivan and Fosso Wamba, 2024) and improving team collaboration through brainstorming and real-time feedback (Jackson et al., 2024), these two types of affordances are also relevant in the GenAI context. However, unlike general technologies, GenAI can generate entirely new content—ranging from text and images to complex simulations and designs (Fosso Wamba et al., 2024; Shore et al., 2024)—thereby giving rise to a novel type of affordance, namely creational affordance (Ramaul et al., 2024).

Although several leading studies have confirmed the role of GenAI in fostering innovation and organizational performance (Wael AL-khatib, 2023; Singh et al., 2024), they typically operationalize GenAI as a single-dimensional construct. Firms, however, can leverage GenAI in multiple ways (Ramaul et al., 2024). This suggests that the impact of GenAI on firms' operations is multifaceted and cannot be adequately captured by a single-dimensional approach. Moreover, competitive intensity denotes the degree of rivalry among firms within a particular industry (Tariq et al., 2023). Although prior studies have examined the moderating role of competitive intensity in the contexts of environmental orientation (Chan et al., 2012), customer orientation (Feng et al., 2019), and innovation (Bachmann et al., 2021), little attention has been devoted to the GenAI context. Intuitively, firms in highly competitive markets are more likely to embrace TO and exploit GenAI affordances to gain or sustain competitive advantage (Bullini Orlandi et al., 2020). However, GenAI is not a universal remedy. This suggests that without a nuanced understanding of how different GenAI affordances function across competitive environments, firms may overestimate the benefits of certain affordances or overlook the potential risks associated with their implementation. To address these research gaps, we pose the following research question: *Do various GenAI affordances mediate the TO–competitive advantage link at different levels of competitive intensity?*

China is a global leader in the adoption and implementation of GenAI (Li et al., 2024). Firms across diverse industries actively employ GenAI to drive product innovation, streamline operations, and enhance customer experiences (Deloitte, 2023). This widespread adoption creates a rich and diverse empirical setting for our study. Moreover, the Chinese market is characterized by intense competition, with numerous firms vying for market share in a rapidly evolving technological landscape (Feng et al., 2019). Thus, this environment offers a particularly valuable context for examining GenAI adoption and its implications. Drawing on survey data from 223 Chinese firms across diverse industries, we contribute to the extant literature on AI-driven business research in three key ways.

First, prior studies on TO have primarily focused on its direct impact on performance outcomes (Chen and Lien, 2013; Bullini Orlandi et al., 2020). By contrast, our study extends this line of inquiry by incorporating GenAI affordances as potential mediators. Specifically, we emphasize that merely recognizing GenAI's technological opportunities is insufficient; to gain or sustain a competitive advantage, firms must effectively integrate and deploy GenAI within their operations.

Second, although traditional technology affordances, such as organizational memory and collaborative affordances, are well established (Chatterjee et al., 2020), GenAI's ability to create entirely new content introduces a novel affordance (Ramaul et al., 2024). By confirming the critical mediating role of GenAI's creational affordance in the TO–competitive advantage relationship, our study offers a more nuanced understanding of how GenAI can be leveraged and extends the application of affordance theory.

Third, although prior studies have examined the moderating role of competitive intensity in various contexts (Chan et al., 2012; Feng et al., 2019), little research has explored this issue in the GenAI context. Our study identifies a unique moderated-mediation effect of competitive intensity on GenAI's creational affordance, in contrast to organizational memory and collaborative affordances. Therefore, our study deepens the understanding of how competitive environments shape technological adoption and strategic outcomes.

2. Literature review

2.1. Technological opportunism

TO refers to the proactive stance of firms in capitalizing on emerging technological developments within dynamic business environments (Bullini Orlandi et al., 2020). Existing research conceptualizes TO as comprising two subdimensions: sensing and responding (Lucia-Palacios et al., 2016). Sensing denotes how organizations detect and interpret technological trends or disruptions, enabling them to anticipate future shifts before they become mainstream (Chen and Lien, 2013). This dimension involves monitoring the external environment and internal R&D activities to identify potential opportunities. By contrast, responding refers to firms' actions in seizing technological opportunities (Lucia-Palacios et al., 2014). Responding encompasses the speed and effectiveness with which organizations adopt, integrate, or adapt new technologies to enhance their capabilities, processes, and products.

Research on TO has predominantly examined its strategic role in enhancing firms' competitive positioning through the exploitation of emerging technologies (Urban and Maphumulo, 2022). Early studies investigated how firms scan their external environments for technological trends and innovations, examining how such foresight facilitates early product development or market entry (Kumar et al., 2025a). Subsequent research explored mechanisms through which firms integrate these opportunities into their operations, including the adoption of new technologies or the modification of business models to leverage emerging advancements (Kanbach et al., 2024). More recent research has extended this inquiry to examine TO's influence on organizational capabilities, such as innovation management and resource allocation, and its potential to yield superior performance outcomes (Liu and Wang, 2024).

Although TO is increasingly recognized as critical for organizational success, the precise mechanisms through which it translates into sustainable competitive advantage remain insufficiently explored. Accordingly, addressing these gaps is essential, as understanding the impact of TO on competitive advantage can offer novel insights into how firms strategically position themselves in increasingly technology-driven and competitive environments. By exploring the underlying mechanisms linking TO to long-term success, future research can yield valuable insights for managers seeking to build strategies that capitalize on technological advancements while mitigating associated risks.

2.2. GenAI affordances

Affordance theory, introduced by psychologist Gibson (1979), was originally developed to explain how individuals perceive the potential uses of objects or resources in their environment and how these interact with their actions and goals. Information systems scholars later extended affordance theory into the domain of technology management (Dincelli and Yayla, 2022). In this context, technology affordances denote the possibilities enabled by technological entities that facilitate users' goal-oriented actions (Anderson and Robey, 2017). The literature identifies organizational memory and collaborative affordances as two primary forms of technology affordances (Chatterjee et al., 2015). Organizational memory affordance refers to how technologies support organizations in storing, reviewing, and retrieving knowledge from their history and experiences. By contrast, collaborative affordance denotes how technology acts as a partner for humans, facilitating cooperation within organizations, particularly in virtual environments.

The scope of technological affordances is broad, encompassing diverse tools ranging from physical devices to software applications. In contrast, as advancements in GenAI continue to reshape industries, growing attention has focused on how firms integrate GenAI into their operations (Kumar et al., 2025b). This shift has prompted an increased focus on GenAI affordances, which denote the unique possibilities for action that GenAI provides to organizations (Ramaul et al., 2024). GenAI's capabilities in natural language processing and machine learning enable the extraction and synthesis of vast amounts of information from diverse sources (Olan et al., 2022), thereby enhancing organizational memory. Moreover, GenAI can proactively assist in brainstorming, ideation, and decision-making (Bouschery et al., 2023), functioning as an intelligent partner that augments rather than merely supports human capabilities. This shift in collaborative dynamics underscores the expanding role of AI in organizational workflows, introducing new ways to coordinate, communicate, and innovate within teams (Ma et al., 2024). Accordingly, organizational memory and collaborative affordances remain applicable in the GenAI context.

However, each new technological paradigm introduces distinct affordances (Ramaul et al., 2024). This is evident in the case of GenAI, which, unlike traditional technologies that primarily automate or support predefined tasks, possesses the capacity to generate entirely new content (Hashmi and Bal, 2024). This unique capacity to generate original outputs has prompted scholars to propose a novel form of affordance—creational affordance—which distinguishes GenAI from general technologies (Li et al., 2025). Therefore, we propose that GenAI affordances comprise three components: organizational memory, collaborative, and creational affordances. Table 1 presents a comparative overview of the three GenAI affordances, highlighting their key roles, main properties, and core foci.

Despite growing scholarly and managerial interest in GenAI affordances, several gaps remain. First, while TO-oriented firms are particularly inclined to recognize and leverage GenAI affordances for competitive advantage (Li et al., 2023a), it is still unclear which specific types of affordances create the greatest value for their operations. This lack of clarity risks inefficient resource allocation, as firms may over-invest in affordances misaligned with their strategic priorities or overlook those with substantial potential benefits. Second, although GenAI affordances are often portrayed as transformative capabilities, their outcomes and associated risks remain insufficiently examined. For instance, as organizations increasingly rely on GenAI to generate novel content, there is a danger that human creativity and judgment could be undermined, resulting in homogenized ideas and reduced innovative capacity (Castro et al., 2025). A deeper investigation into both the benefits and unintended consequences of GenAI affordances is therefore critical to understanding how firms can exploit these technologies responsibly.

2.3. Competitive intensity

Competitive intensity reflects the degree of rivalry within an industry or market, shaped by factors such as the number of competitors and the magnitude of competitive forces (Meena et al., 2024). Existing literature predominantly examines its role in shaping firm behavior and performance (Olabode et al., 2022). For instance, Lei et al. (2024) analyze survey data from 325 firms and show that in highly competitive environments, firms tend to share both tacit and explicit knowledge to foster frugal innovation capabilities. Similarly, Crick et al. (2024) analyze data from 262 U.S. firms and demonstrate that competitive intensity moderates the relationship between coopetition strategies and performance outcomes. Although these studies provide valuable insights into how firms adapt to competitive pressures (Su et al., 2024), the broader dynamics of competitive intensity, particularly in the context of emerging technological advancements, remain underexplored.

Affordances, as prior research suggests, depend largely on how users engage with them to achieve specific goals (Ramaul et al., 2024). This perspective implies that competitive intensity may influence both the speed and manner in which TO-oriented firms integrate GenAI into their strategic activities. For example, in highly competitive environments, such firms may be compelled to adopt GenAI aggressively to achieve differentiation, leveraging its creational affordance to develop novel products or services. Conversely, in less competitive markets, the drive to innovate may be slower, with TO-oriented firms placing greater emphasis on operational efficiency, such as strengthening organizational memory and utilizing collaborative affordances to streamline internal processes. Understanding the moderating effect of competitive intensity is therefore crucial for analyzing how TO, through GenAI affordances, shapes competitive advantage. This interplay ultimately determines the strategic success of technological adoption.

3. Hypothesis development

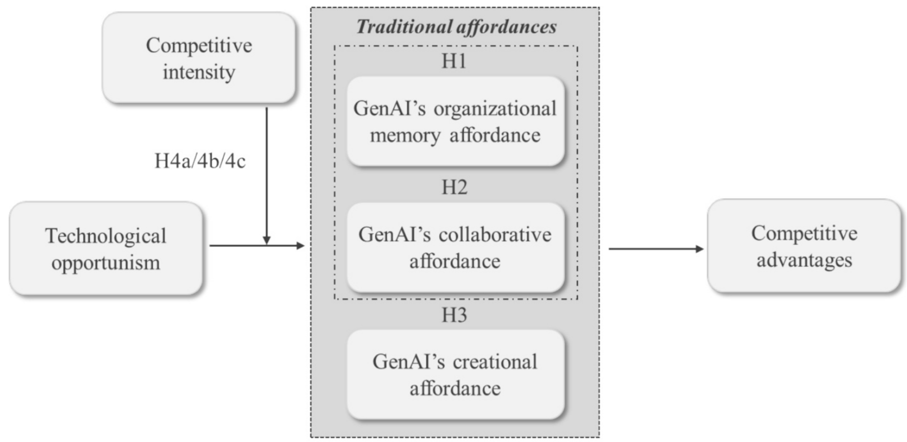
TO underscores the importance of firms adopting a proactive stance to promptly identify external technological shifts and adjust their strategies to capitalize on them (Li et al., 2023a). Recent evidence suggests that 71 % of organizations failing to keep pace with technological advancements risk falling behind competitors (Capgemini Research Institute, 2024). This underscores the critical role of TO in determining whether firms can cultivate competitive advantage. However, some scholars contend that although TO reflects firms' awareness of the potential value of new technologies and their willingness to adopt them, this orientation does not necessarily translate into tangible competitive benefits (Ramaul et al., 2024). This occurs because organizations may adopt the same technology yet implement it differently, leading to divergent outcomes (Gibson et al., 2022). Accordingly, firms' ability to gain a competitive advantage depends not only on recognizing technological opportunities but also on effectively leveraging the affordances offered by technologies such as GenAI. TO explains why firms are motivated to adopt GenAI, whereas GenAI affordances specify how firms mobilize, orchestrate, and apply the capabilities of GenAI technologies. We therefore ground our analysis in affordance theory to examine how GenAI affordances influence the TO–competitive advantage relationship and how competitive intensity moderates these dynamics. Fig. 1 presents the conceptual model.

3.1. Mediating role of GenAI affordances

Our study examines whether different GenAI affordances mediate the relationship between TO and competitive advantage under varying levels of competitive intensity, rather than exploring the interrelationships among the three affordances. Accordingly, the three affordances

Table 1
Differences between the three dimensions of GenAI affordances.

	Organizational memory affordance	Collaborative affordance	Creational affordance
Key roles	Accumulation and retrieval of past experiences; enhancing knowledge inheritance and accessibility	Support for multiple rounds of dialogue; enhancing collaborative efficiency and real-time communication	Stimulating inspiration and creativity, generating novel ideas, and inspiring innovation
Main properties	Stability, integrity, and traceability of memory	Real-time nature, synergy, and transparency of communication	Novelty, diversity, and openness of the output
Core foci	Emphasizes knowledge reproduction rather than novelty	Focuses on the interaction process	Emphasizes breaking conventions and creating new content



Note: H1, H2, and H3 predict mediation of the relationship between technological opportunism and competitive advantages, with organizational memory affordance, collaborative affordance, and creational affordance as the respective mediators. H4a, H4b, and H4c predict moderation of the same mediation relationships by competitive intensity.

Fig. 1. Conceptual model.

are conceptualized as parallel mediators, allowing us to assess their individual effects without assuming any directional or hierarchical interdependencies.

Affordance theory posits that technologies provide certain opportunities for action based on their inherent capabilities (Anderson and Robey, 2017). Because TO reflects a proactive and adaptive mindset (Malik et al., 2024), firms with strong TO are particularly likely to recognize the potential of GenAI in enhancing organizational memory. Specifically, GenAI can process, analyze, and synthesize vast amounts of data, enabling the development of knowledge repositories that are comprehensive and dynamic (Retkowsky et al., 2024). By leveraging these capabilities, TO-oriented firms can build substantially more robust organizational memory, drawing insights from both internal and external sources (Olan et al., 2022). For instance, Airbus applies GenAI to process technical data in real time, thereby creating adaptive, knowledge-rich systems that allow the company to rapidly integrate new insights into decision-making and maintain an evolving organizational memory (Perera, 2024).

The outcomes of GenAI's organizational memory affordance are particularly valuable. Prior studies demonstrate that by rapidly accumulating and integrating large volumes of data, organizational memory enables organizations to access and leverage insights from both historical and real-time sources, thereby accelerating response times and improving overall efficiency (Gibson et al., 2022). Moreover, by synthesizing data across domains, firms can identify emerging trends, optimize existing processes, and develop innovative solutions more effectively than competitors lacking such capabilities (Chatterjee et al., 2015). Consistent with affordance theory, while TO allows firms to recognize the potential value of technologies such as GenAI, sustainable competitive advantage can only be realized when these opportunities are effectively translated into action.

H1. GenAI's organizational memory affordance mediates the TO–competitive advantage link.

TO-oriented firms are more likely to exhibit a strong collaborative affordance of GenAI. Specifically, such firms prioritize cultivating a culture of innovation by emphasizing exploration, experimentation, and the continuous integration of advanced technologies (Chen and Lien, 2013). As an emerging technology, GenAI enables organizations to redefine the boundaries of human–AI collaboration: employees contribute professional judgment and critical thinking, while AI provides speed, scalability, and quantitative capabilities (Wilson and Daugherty, 2018). The complementary strengths of AI and human workers further enhance cross-departmental collaboration, breaking down traditional silos and promoting highly efficient workflows (Sundberg and Holmström, 2023; Ramaul et al., 2024). A notable example can be seen at Rockwell Automation, a leading industrial manufacturer, which leverages GenAI to assist engineers in generating code. This approach optimizes development processes and facilitates seamless communication among departments, including engineering, IT, and product design (Blackman, 2023).

With respect to outcomes, prior research indicates that collaborative affordance facilitates real-time interaction and feedback among team members, supporting effective information sharing, rapid problem-solving, and improved decision-making (Chatterjee et al., 2015). In addition, collaborative affordance enables the integration of diverse perspectives, fostering the creation of innovative solutions and promoting a highly adaptive organizational culture (Gibson et al., 2022). Overall, TO-oriented firms are likely to recognize and exploit the collaborative affordance of GenAI because it allows them to respond rapidly to market changes and deliver novel solutions, thereby enhancing competitive advantage.

H2. GenAI's collaborative affordance mediates the TO–competitive advantage link.

Organizations engaging in TO are particularly likely to recognize the creational affordance of GenAI. TO typically requires a high tolerance for risk and a strong commitment to innovation (Urban and Maphumulo, 2022). Because GenAI enables the generation of novel ideas, concepts, and solutions that may not be achievable with traditional technologies (Ramaul et al., 2024), TO-oriented organizations are inclined to experiment with these creative potentials, understanding that the potential rewards of innovation often outweigh associated risks. Furthermore, TO-driven firms tend to allocate resources toward exploring emerging technologies that may not yet have a proven track record but hold substantial potential (Srinivasan et al., 2002). GenAI's creational affordance aligns closely with this approach, as firms recognize its capacity to unlock new creative opportunities even when its full potential is not yet fully understood. For instance, Amgen, a U.S. biopharmaceutical company, has successfully integrated GenAI into its antibody design process, illustrating the practical benefits of leveraging this affordance (ArdorComm News Network, 2024).

Regarding outcomes, prior studies indicate that GenAI's creational affordance accelerates the development of novel products and services, enabling firms to stay ahead of market trends and differentiate themselves from competitors (Li et al., 2025; Bouschery et al., 2023; Retkowsky et al., 2024). Leading firms such as the Tata Group have utilized GenAI to enhance conversational shopping experiences and personalize marketing campaigns (Bardia, 2024). Beyond product and service innovation, creational affordance can generate operational efficiencies: by automating creative processes, firms can free human resources for more strategic, high-value tasks while improving output quality and consistency (Ramaul et al., 2024; Holmström and Carroll, 2024). Overall, because GenAI's creational affordance supports rapid idea generation, market responsiveness, and enhanced operational efficiency, TO-oriented firms are likely to identify and exploit these affordances.

H3. GenAI's creational affordance mediates the TO–competitive advantage link.

3.2. Moderating role of competitive intensity

From the perspective of affordance theory, technological affordances are neither inherent nor universally accessible features of a technology (Ramaul et al., 2024). Instead, they emerge through a relational perception process, whereby users interpret and enact technological features based on their goals, motivations, and contextual demands (Chatterjee et al., 2020). Given that competitive intensity reflects the degree of market dynamism, customer pressure, and rival actions that firms must navigate (Olabode et al., 2022; Meena et al., 2024), it is likely to significantly influence how technological features are perceived and utilized.

First, TO-oriented firms operating in highly competitive environments are particularly attuned to leveraging emerging technologies, such as GenAI, to expand their knowledge boundaries (Kinkel et al., 2022). This is primarily because the domain knowledge advantage provided by GenAI enables more comprehensive strategic analyses, presenting multiple actionable options for decision-making (Gangwar et al., 2025). By storing, retrieving, and analyzing historical data and organizational knowledge, GenAI's organizational memory affordance allows these firms to build robust knowledge repositories, rapidly extract insights, identify potential technological trends or market gaps, and optimize future technology and product development based on past experiences (Chatterjee et al., 2015). A notable example is Amazon, which leveraged AI to analyze browsing behavior, purchase history, and visual preferences, thereby increasing sales by 35 % through personalized recommendations and enhancing customer loyalty by 20 % (Kamal et al., 2024).

Second, GenAI's collaborative affordance is particularly critical for TO-oriented firms in highly competitive environments, where efficient human–AI collaboration is essential for tackling complex tasks. GenAI fundamentally reshapes human–technology interactions (Li et al., 2023b). Firms must possess the capabilities to collaboratively utilize GenAI in strategy formulation, enabling them to identify and implement the most promising approaches to enhance competitive advantage. Rather than merely functioning as a tool, GenAI acts as a collaborative partner within the decision-making process (Ramaul et al., 2024). In highly competitive markets, TO-oriented firms are inclined to become "technology collaborators," leveraging collaborative affordance to coordinate the efforts of multiple employees and AI. For instance, BMW integrates AI not to replace human expertise but to complement it in developing autonomous driving systems. AI assists in analyzing vast datasets, accelerating technological development, and enhancing safety and precision, thereby strengthening BMW's competitive position (Aicadium, 2023).

Third, firms in highly competitive environments face pressure from rapid product updates and market shifts, making technological tools essential for accelerating innovation. GenAI's creational affordance equips firms with novel ideas and solutions in product design, marketing strategies, and other domains (Ramaul et al., 2024). Accordingly, a highly competitive environment motivates TO-oriented firms to actively explore GenAI's creative potential to stimulate innovation, develop distinctive value propositions, and launch new products, services, or processes. For example, ByteDance, amid intense competition in the short-video platform market, leveraged its AI-powered recommendation engine (BytePlus) to rapidly iterate its content recommendation system, significantly enhancing user experience (Phan, 2021). Similarly, Haier, operating in the smart home sector, employs AI technology to power its "Home GPT" solutions, enabling the rapid launch of personalized products that attract consumers in a highly competitive market (IoT EXPO, 2023).

In summary, we posit that firms operating under high competitive intensity face strong pressures to innovate rapidly, differentiate their offerings, and optimize resource allocation. Such external pressures increase their sensitivity to GenAI affordances and strengthen their motivation to exploit them. Conversely, in low-competition environments, the same level of TO may not lead to proactive engagement with GenAI affordances, as the urgency and perceived strategic value of such actions are reduced. Therefore, competitive intensity functions as a contextual amplifier, translating firms' internal disposition (i.e., TO) into observable affordance realization. Considering these dynamics, along with the mediating role of GenAI affordances in the relationship between TO and competitive advantage, we propose the following moderated-mediation hypotheses:

H4a. Competitive intensity positively moderates the relationship between TO and GenAI's organizational memory affordance, thereby strengthening the indirect effect of TO on competitive advantage through organizational memory affordance.

H4b. Competitive intensity positively moderates the relationship between TO and GenAI's collaborative affordance, thereby strengthening the indirect effect of TO on competitive advantage through collaborative affordance.

H4c. Competitive intensity positively moderates the relationship between TO and GenAI's creational affordance, thereby strengthening the indirect effect of TO on competitive advantage through creational affordance.

4. Research design

4.1. Survey instrument

Several factors motivated our focus on Chinese firms. First, as the world's most populous country undergoing rapid digital transformation, China presents a vast consumer market with growing demand for innovative products and services. This demand encourages firms to seize technological opportunities, incorporating GenAI into product development, service enhancement, and operational optimization. Second, numerous Chinese IT firms, such as DeepSeek and Baidu, have made substantial investments in GenAI, achieving rapid advancements and tangible results. These firms provide valuable insights into the latest developments and real-world applications of GenAI. Finally, GenAI adoption in China extends beyond the IT sector, with widespread use across industries including manufacturing, e-commerce, finance, and healthcare. This broad adoption allows for a comprehensive examination of how TO can be leveraged to achieve competitive advantages through GenAI affordances under varying levels of competitive intensity.

We designed a structured questionnaire with careful attention to clarity and neutrality. To pre-test the instrument, 50 MBA students were invited to assess the survey for complexity, ambiguity, and leading questions. In the formal data collection, a cover letter accompanied the questionnaire to ensure respondents consistently understood the items. The cover letter served three purposes: (1) providing respondents with an overview of the research context, (2) offering clear explanations of key constructs (e.g., GenAI) to minimize ambiguity, and (3) emphasizing that responses should reflect the organization's overall application of GenAI rather than specific use cases or task-level outcomes. This approach ensured alignment between the survey items and the intended level of analysis.

To guarantee that respondents had practical experience with GenAI, we incorporated screening questions, such as "Has your company integrated GenAI into its business operations?" and "How much has your company invested in GenAI-related projects over the past year?" Respondents lacking familiarity with or direct involvement in GenAI were excluded, thereby enhancing sample relevance and validity. We deliberately did not include firms that had not adopted GenAI for two reasons: (1) the study aims to examine how GenAI affordances influence the TO–competitive advantage relationship, not to compare users and non-users; (2) non-adopting firms generally lack awareness of GenAI affordances and cannot provide meaningful responses to the survey items.

Data collection was facilitated through a well-established online survey platform with access to a professional network of over 30,000 Chinese companies. This platform enabled targeted outreach to GenAI-using firms across diverse industries. To encourage participation, respondents received an e-gift card and were entered into a lottery for a cash reward. To ensure data quality, multiple control measures were implemented: (1) surveys with continuous identical answers were excluded to remove careless responses; (2) only responses from senior-level employees were retained to ensure expertise; (3) duplicate IP addresses were removed to prevent redundancy; (4) surveys completed in under five minutes or over 15 min were excluded as likely reflecting rushed or excessively prolonged responses. Following these procedures, 223 valid responses were obtained from 365 distributed surveys, yielding a response rate of approximately 61.10 %. Descriptive statistics for the sample are provided in Table 2.

4.2. Measures

TO, GenAI affordances, competitive intensity, and competitive advantages are the main constructs examined in the present study. To assess TO, we followed Srinivasan et al. (2002) and used four items. GenAI affordances comprise three components: organizational memory affordance, collaborative affordance, and creational affordance. Measurements for the first two components were adapted from Chatterjee et al. (2020), while those for the third component were adapted from Li et al. (2025). Five items were derived from the scale developed by Jaworski and Kohli (1993) to evaluate competitive intensity. Competitive advantage was measured using four items reflecting financial and strategic performance, consistent with prior research (Schilke, 2014; Li et al., 2023c). Financial performance—determined through such indicators as EBIT, ROI, and ROS—reflects firms' short-term efficiency and profitability relative to industry averages. By contrast, strategic performance—measured by market share—represents firms' long-term market position and ability to sustain their competitive edge over current and potential rivals. A 7-point Likert scale was used in this study to determine the respondents' judgments. Fig. A1 in the Appendix lists all items used to measure each construct. Lastly, four firm characteristics mentioned in Table 1 were operated as control variables.

Table 2
Demographic profile.

Firm information	Obs.	%
Number of employees		
≤299	80	35.87
300–999	102	45.74
≥1000	41	18.39
Number of years in operation		
<6 years	5	2.24
6–10 years	44	19.73
11 − 15 years	69	30.94
16 − 20 years	65	29.15
>20 years	40	17.94
Ownership		
State-owned	18	8.07
Privately owned	178	79.82
Others	27	12.11
Industry types		
Manufacturing	87	39.01
IT industry	116	52.02
Others	20	8.97

5. Data analysis and results

5.1. Measurement model assessment

First, Cronbach's α and composite reliability (CR) values for the variables ranged from 0.801 to 0.884 and 0.802 to 0.887, respectively, with all values exceeding the threshold of 0.7 (see Fig. A1 in the Appendix), thereby demonstrating strong reliability (Li et al., 2022). To establish content validity, all measurement items were adapted from well-established instruments used in previous studies. In addition, a small-scale pilot test was conducted prior to the main survey to gather feedback from informants. We utilized CFA to assess convergent validity. The results (χ^2 = 331.867, df= 237, χ^2/df= 1.400, TLI = 0.960, CFI = 0.965, IFI = 0.966, and RMSEA = 0.042) indicate a good model fit. Moreover, all measurement items exhibited significant factor loadings ranging from 0.653 to 0.885 (see Fig. A1 in the Appendix), thereby confirming the scale's convergent validity. Lastly, discriminant validity was evaluated by comparing the square root of the average variance extracted (i.e., $\sqrt{AVE}$) with its correlations to other constructs. The results show that $\sqrt{AVE}$ exceeded the correlation coefficients (see Table 3), indicating satisfactory discriminant validity.

5.2. Bias statements

We utilized several strategies to mitigate potential biases in this study. First, we used the early vs. late respondents' comparison method to assess non-response bias. Respondents were categorized as "early respondents" (first quarter) and "late respondents" (last quarter) based on their response times. Levene's test was utilized to examine variance differences in key variables, followed by an independent samples *t*-test to assess mean differences (Marodin et al., 2018). The results indicate no significant differences between the two groups, suggesting that non-response bias was not a significant concern.

Second, common method bias (CMB) refers to the systematic inflation or distortion of correlations between variables owing to a shared data source, measurement method, or environment. To mitigate CMB, we emphasized anonymity and confidentiality to reduce social desirability. Furthermore, after data collection, we performed Harman's single-factor test by loading all measurement variables into a single factor. The results reveal poor model fit (χ^2 = 1054.477, df = 252, χ^2/df = 4.184, TLI = 0.680, CFI = 0.708, IFI = 0.711, and RMSEA = 0.120), indicating that the method factor had low explanatory power. In addition, a 7-point Likert scale was utilized to assess respondents' family well-being, serving as the marker variable in this study. Significant correlations between the marker variable and other variables generally indicated that CMB is severe (Williams and O'Boyle, 2015). However, the results reveal that the marker variable was uncorrelated with the other variables (see Table 2), further supporting the conclusion that CMB did not pose a significant issue in our study.

5.3. Hypothesis testing

Studies have shown that strong correlations among independent variables can lead to multicollinearity, possibly resulting in biased estimates of coefficients, inflated standard errors, and overall model instability (Kyriazos and Poga, 2023). To address multicollinearity and ensure the validity of the analysis, we calculated the variance inflation factor (VIF) values for each independent variable in the model. Upon analysis, all VIF values in our model were below the threshold of 5 (Li et al., 2024), with the highest being 1.872. Hence, multicollinearity is not a major concern. To test our research hypotheses, we utilized the PROCESS macro, a robust tool for mediation and moderation analysis, within the SPSS software (Hayes, 2015; Li et al., 2024). We specifically utilized the default Model 4 to test whether the relationship between TO and competitive advantages is mediated by GenAI affordances. In

Table 3
Descriptive statistics and correlations.

	1	2	3	4	5	6
1. Technological opportunism	**0.768**					
2. Organizational memory affordance	0.583^{**}	**0.739**				
3. Collaborative affordance	0.525^{**}	0.580^{**}	**0.757**			
4. Creational affordance	0.586^{**}	0.535^{**}	0.501^{**}	**0.758**		
5. Competitive intensity	0.561^{**}	0.417^{**}	0.372^{**}	0.428^{**}	**0.782**	
6. Competitive advantages	0.639^{**}	0.570^{**}	0.526^{**}	0.558^{**}	0.415^{**}	**0.729**
7. Marker variable	0.088	0.124	−0.019	−0.007	0.047	0.062
Mean	5.751	5.842	5.827	5.830	5.405	5.797
Standard deviation	0.633	0.580	0.675	0.620	0.813	0.578

Notes: ** indicates p-values below 0.01; the diagonal contains $\sqrt{AVE}$.

addition, we used the default Model 7 to test whether the relationship between TO and GenAI affordances varies across different levels of competitive intensity and further examined how these effects translate into differences in competitive advantage.

The PROCESS macro first reports the results based on the regression analysis. As shown in model 7 of Table 4, we entered the control variables along with TO's effect on competitive advantages. Without any mediators, the direct effect of TO on competitive advantages is positive and significant ($\beta = 0.588$; $p < 0.001$). For models 1, 3, and 5, we sequentially entered the control variables, along with TO's effect on each of the three mediator variables, separately. TO exhibits positive relationships with the three types of GenAI affordances: organizational memory affordance ($\beta = 0.532$; $p < 0.001$), collaborative affordance ($\beta = 0.556$; $p < 0.001$), and creational affordance ($\beta = 0.571$; $p < 0.001$). For model 8, we simultaneously incorporated TO and the three mediators (along with the control variables). TO and the three mediators demonstrate significant associations with competitive advantages (TO: $\beta = 0.314$, $p < 0.001$; organizational memory affordance: $\beta = 0.186$, $p < 0.01$; collaborative affordance: $\beta = 0.132$, $p < 0.05$; creational affordance: $\beta = 0.177$, $p < 0.01$).

Then, we constructed the interaction between TO and competitive intensity (TO × CI) and conducted a moderated mediation analysis to examine whether or not the indirect effects were contingent on competitive intensity. As shown in models 2, 4, and 6 of Table 4, TO × CI shows no significant associations with organizational memory affordance ($\beta = 0.001$; $p > 0.05$) or collaborative affordance ($\beta = -0.023$; $p > 0.05$) but reveals a significantly positive association with creational affordance ($\beta = 0.211$; $p < 0.01$).

Results from the PROCESS macro, which utilizes the bootstrapping approach, are also provided. As shown in Table 5, the three types of GenAI affordances mediate the relationship between TO and competitive advantages, with 95 % confidence intervals excluding zero. Furthermore, we analyzed the moderated-mediation indices for GenAI affordances at varying levels of competitive intensity. Note that only the indices for creational affordance were significant (index = 0.037, SE = 0.019; 95 % CI [0.006, 0.080]). These effect sizes are meaningful for two reasons. First, the mediation effects of organizational memory and creational affordances are slightly larger than that of collaborative affordance, suggesting that while all three affordances significantly translate TO into competitive advantages, creational and organizational memory affordances exert relatively stronger influence. Second, the moderated-mediation effect size indicates that under higher competitive intensity, the contribution of creational affordance to competitive advantage becomes more pronounced, underscoring its practical relevance in highly dynamic environments. Accordingly, H1, H2, H3, and H4c are

Table 5
Results of the bootstrapping approach.

Indirect paths	Effect	BootSE	BootLLCI	BootULCI
Organizational memory affordance	0.099	0.047	0.016	0.200
Collaborative affordance	0.074	0.037	0.014	0.159
Creational affordance	0.101	0.039	0.026	0.181
Total	0.274	0.057	0.180	0.398

Table 4
Results of the regression analysis.

	Organizational memory affordance		Collaborative affordance		Creational affordance		Competitive advantages	
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Constant	2.709^{***}	2.592	2.715^{***}	1.849	2.532^{***}	8.933^{***}	2.549^{***}	1.237^{***}
Control variables								
Firm age	0.032	0.030	−0.064	−0.066	−0.013	−0.015	0.008	0.013
Firm size	−0.019	−0.016	0.130*	0.132*	0.072	0.077	−0.066	−0.092
State-owned	−0.093	−0.058	−0.068	−0.030	−0.099	−0.066	0.077	0.121
Privately owned	−0.107	−0.099	0.020	0.028	−0.027	−0.018	0.008	0.030
IT industry	0.135	0.143	−0.105	−0.098	−0.070	−0.049	−0.075	−0.074
Manufacturing	0.057	0.069	−0.160	−0.148	−0.029	−0.017	−0.042	−0.026
Independent variable								
Technological opportunism (TO)	0.532^{***}	0.462	0.556^{***}	0.614	0.571^{***}	−0.648	0.588^{***}	0.314^{***}
Mediating variables								
Organizational memory affordance								0.186^{**}
Collaborative affordance								0.132*
Creational affordance								0.177^{**}
Moderating variables								
Competitive intensity (CI)		0.090		0.232		−1.115*		
TO × CI		0.001		−0.023		0.211^{**}		
Degrees of Freedom	7, 215	9, 213	7, 215	9, 213	7, 215	9, 213	7, 215	10, 212
R^2	0.352	0.363	0.294	0.303	0.351	0.389	0.415	0.519
F value	16.651^{***}	13.485^{***}	12.798^{***}	10.303^{***}	16.635^{***}	15.076^{***}	21.823^{***}	22.891^{***}

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

supported, but H4a and H4b are rejected.

6. Discussion

This section summarizes the core findings of our study and situates them within the existing literature. Prior research suggests that firms with high levels of TO are more likely to achieve superior performance outcomes compared with firms adopting a wait-and-see approach (Li et al., 2023a; Bullini Orlandi et al., 2020). However, in practice, not all organizations can effectively leverage emerging technologies, such as GenAI, even after recognizing their potential. Our study indicates that without the effective activation of GenAI affordances, firms may fall into a "technology trap.".

Drawing on the affordance perspective, this study identifies a key mechanism through which TO influences competitive advantage: the shaping of GenAI affordances in organizational memory, collaboration, and creativity. First, after identifying technological opportunities, organizations must integrate insights into organizational memory, update knowledge maps, and enhance adaptability. This integration enables rapid translation of data into actionable strategic decisions, reducing the lag between information acquisition and execution (Cegarra-Navarro and Martelo-Landroguez, 2020). Second, TO generates value by enhancing human–AI collaboration. GenAI fosters communication, transparency, and feedback, dismantling organizational barriers and supporting team collaboration, which in turn promotes organizational performance and competitive advantages (Sundberg and Holmström, 2023; Ramaul et al., 2024). Third, the creational affordance of GenAI directly bolsters innovation (Ramaul et al., 2024). It allows organizations to approach problems from multiple perspectives, generate new concepts, and propose transformative solutions, thereby expanding the scope of creativity and enabling innovative business processes. These mediating findings align with Chatterjee et al. (2020), who emphasize that IT affordances are shaped not only by the technology itself but also by how organizations actively integrate them into workflows. Furthermore, the continual adaptation and learning from technology-driven insights create a feedback loop that reinforces organizational innovation capacity.

An intriguing finding is that among the three affordance types, only creational affordance exhibits an enhanced mediating effect under high competitive intensity. A plausible explanation is that in highly competitive environments, organizations must rapidly adapt to changes, and creational affordance directly supports innovation and adaptability (Ramaul et al., 2024). Additionally, creational affordance addresses the challenge of "generalizability," i.e., the consistency of AI model performance across diverse domains. While AI systems with strong generalizability are versatile, they may lack the flexibility required for dynamic adjustments. Cocreational affordance, by contrast, encourages exploration beyond conventional solutions, enabling AI systems to adapt to changing environments in a manner akin to living organisms (Ramaul et al., 2024). This aligns with Sundberg and Holmström (2024), who note that creational affordance allows organizations to generate and combine different types of content, exploring novel combinations that constitute the true advantage of GenAI.

Finally, the absence of significant moderated-mediation effects of competitive intensity on GenAI's organizational memory (H4a) and collaborative affordances (H4b) can be attributed to their relatively stable and internally embedded nature. Under intense competition, managerial attention and organizational resources are typically directed toward capabilities offering immediate, visible gains. Creational affordance, which enables rapid idea generation and innovation, aligns with firms' pressing need for differentiation, becoming a focal point of strategic action (Li et al., 2025). In contrast, organizational memory affordance, which supports decision-making through the retrieval and analysis of historical knowledge, is often regarded as a background capability (Argote, 2013). While it contributes to long-term learning and process optimization, it lacks the urgency that commands executive attention during competitive crises. Similarly, collaborative affordance, although critical for integrating human and AI efforts, functions as a baseline requirement in the digital era. Its strategic importance remains steady across contexts but does not fluctuate significantly with competitive intensity. Consequently, organizational memory and collaborative affordances primarily operate as enabling infrastructures rather than context-sensitive strategic levers.

6.1. *Theoretical implications*

Our study offers significant contributions to the existing body of AI-driven business literature. First, the extant research on TO has mainly concentrated on its direct effect on organizational performance outcomes (Malik et al., 2024; Chen and Lien, 2013; Bullini Orlandi et al., 2020). Our study extends this body of work by introducing a markedly nuanced layer of analysis. We specifically draw upon affordance theory and propose that merely recognizing GenAI as a technological opportunity is insufficient for achieving a sustainable competitive advantage. To fully leverage the potential of GenAI, firms must go beyond merely identifying opportunities; they must effectively integrate and deploy GenAI across their operations, processes, and strategies. By framing the affordances of GenAI as central to this process, we provide a substantially comprehensive understanding of how firms can utilize technological opportunities to achieve and sustain a competitive advantage.

Second, although traditional technology affordances, such as organizational memory and collaborative affordances, have been well-established in the literature (Chatterjee et al., 2015; Chatterjee et al., 2020), the advent of GenAI introduces a novel dimension to this framework. In particular, GenAI's ability to generate entirely new content introduces a new type of affordance—creational affordance (Ramaul et al., 2024)—which expands the boundaries of what technology can achieve for organizations. By revealing how the three distinct types of GenAI affordances mediate the relationship between TO and competitive advantages, our study offers a profound insight into how firms can strategically harness GenAI. Moreover, the introduction of creational affordance challenges traditional notions of technology's role in organizations and opens new avenues for exploring how different affordances interact to produce strategic outcomes.

Third, although prior literature has examined the influence of competitive intensity on firm behavior and performance (Crick et al., 2024; Lei et al., 2024), limited attention has been given to how this factor influences firms' adoption and usage of emerging technologies, particularly in highly dynamic business environments. The current study extends existing knowledge by demonstrating how competitive intensity moderates the relationship between TO and competitive advantages through GenAI affordances. By contrasting creational affordances with organizational memory and collaborative affordances, we highlight the differential effects of competitive intensity on how firms use GenAI across different market conditions. This novel approach enriches the understanding of competitive intensity's role in technology adoption and contributes to the broad literature on the strategic implications of emerging technologies.

6.2. *Practical implications*

Our study provides actionable guidance for practitioners seeking to integrate GenAI effectively into their strategic frameworks. First, to successfully leverage GenAI and attain a competitive advantage, firms must adopt a forward-looking mindset regarding TO. This undertaking involves recognizing the potential of GenAI and possessing the capability to fully explore and capitalize on its affordances. Although GenAI technologies offer diverse affordances, organizations often fail to utilize them effectively, leading to suboptimal outcomes or even strategic misalignments. In practice, some firms may over-invest in ill-suited affordances that do not align with their strategic needs. A typical example is IBM's investment in Watson for cancer treatment recommendations. Despite Watson's powerful organizational memory affordance, the highly complex practical context in the medical industry and doctors' requirements for "interpretability and trust" were overlooked. Consequently, the system frequently offered treatment suggestions that were either inaccurate or clinically unrealistic, ultimately leading to the failure of its medical application (Dolfing, 2024). Therefore, firms should focus on understanding and leveraging GenAI's affordances, adapting the technology to meet specific business needs. Moreover, GenAI adoption should not follow a one-size-fits-all approach. Instead, it must be deeply integrated into the organization's unique requirements and capabilities.

Second, firms in highly competitive environments must prioritize the development and exploitation of GenAI's creational affordances to sustain innovation and differentiation. Competitive intensity creates external pressure that forces firms to seek novel ideas, faster design iterations, and considerably personalized solutions. GenAI, when used strategically, can act as a creative partner—accelerating ideation, offering diverse design alternatives, and fostering rapid intellectual exchange within teams. However, this potential can only be fully realized when human creativity and GenAI capabilities are effectively integrated. The case of Balenciaga serves as a cautionary example: although the brand adopted GenAI in its creative processes, critics raised concerns about the dilution of human-centered creativity and brand identity. As Edoardo Tocco, President of Balenciaga Asia Pacific, emphasized, "AI inspires designers' creativity, but final decisions and aesthetic judgments remain the designer's responsibility" (Girod, 2024). The missed opportunity was not the use of AI itself but the insufficient cultivation of designer-AI complementarity. This aspect highlights a markedly broad managerial implication—firms must actively foster a culture that encourages experimentation, provides design teams with flexible AI tools, and ensures human oversight in decision-making. Only then can GenAI's creational affordance translate into tangible competitive gains in a fast-moving market.

Third, firms should not overlook the enduring strategic value of organizational memory and collaborative affordances, even if their impact appears minimally influenced by competitive intensity. These affordances operate as foundational enablers rather than reactive differentiators. Organizational memory affordance enables firms to embed GenAI-generated content within historical knowledge structures—improving learning continuity and long-term decision-making. This affordance type serves as the backbone of sustained advantage, enabling GenAI outputs to inform future strategies, refine operations, and support cross-cultural organizational learning. Similarly, collaborative affordance ensures that GenAI's value is not confined to siloed teams but is shared, debated, and integrated across departments. In creative fields, such as advertising and product design, GenAI can spark team-level innovation, but this outcome depends on strong internal collaboration, psychological safety, and a shared understanding of AI tools. Even in industries with relatively low competitive pressure, collaboration is essential for building AI readiness and embedding GenAI into everyday workflows. Therefore, although creational affordance should be emphasized under competitive pressure, firms must simultaneously invest in building robust organizational memory systems and cultivating a collaborative environment to unlock GenAI's full, long-term potential.

7. Conclusion and future research

We develop a new framework grounded in affordance theory to help forward-thinking firms effectively leverage the potential of GenAI to enhance competitive advantage. We explore the relationships among TO, three types of GenAI affordances, and competitive advantage across varying levels of competitive intensity. Through a survey of 223 Chinese firms, we confirm the significant mediating role of organizational memory, collaborative, and creational affordances in the TO–competitive advantage relationship. Moreover, we identify the unique moderating–mediating effect of competitive intensity on creational affordance. Our research contributes to the existing AI-enabled business literature by highlighting the mechanisms and boundary conditions through which firms with a proactive stance toward emerging technologies can shape competitive advantage. Our findings also provide firms with detailed guidance on optimizing the strategic deployment of GenAI.

Despite our considerable effort, several aspects warrant further investigation. First, although we confirm the positive mediating effect of GenAI affordances on the TO–competitive advantage link, this aspect is far from being an exhaustive picture. TO firms can gain a highly competitive advantage through many different mechanisms. Beyond the affordance theory perspective, future research may explore alternative theoretical lenses to reinterpret and further dissect the mechanisms through which TO influences competitive advantage.

Second, although the three GenAI affordances are treated as parallel mediating mechanisms in this study, they do not function in isolation in practice and may follow a sequential logic. For example, organizational memory affordance may provide the knowledge foundation that enhances collaborative affordance, which, in turn, could stimulate creational affordance by facilitating substantially extensive idea exchange and co-creation. Therefore, future research could investigate the potential interdependencies among these affordances.

Lastly, our study does not address the potential side effects of GenAI, such as disruptions to knowledge ties, labor displacement, and algorithmic bias (Retkowsky et al., 2024). These issues could undermine the positive outcomes associated with GenAI. Future research could investigate the role of organizational ethical capabilities, providing a superior understanding of how to mitigate the ethical risks of GenAI deployment.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT in order to improve the readability and language of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

CRediT authorship contribution statement

Lixu Li: Writing – original draft, Funding acquisition, Formal analysis, Conceptualization. **Mengchao Wu:** Writing – original draft, Funding acquisition. **Yaoqi Liu:** Writing – review & editing, Supervision, Funding acquisition. **Yong Jin:** Writing – review & editing, Supervision. **Qiang Li:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Construct	Items
Technological opportunism (CR = 0.851; AVE = 0.590; Cronbach's α = 0.847)	• We actively detect technological developments that may potentially affect our business (0.798). • We actively seek intelligence on technological changes in the environment that are likely to affect our business (0.759). • We periodically review the likely effect of changes in technology on our business (0.832). • We generally respond very quickly to technological changes in the environment (0.675).
Organizational memory affordance (CR = 0.827; AVE = 0.546; Cronbach's α = 0.827)	*In our company, GenAI is used to...* • ...compile information about work tasks (0.771). • ...analyze historical information about business operations (0.681). • ... retrieve, share, and reuse best practice information (0.731). • ... create an online knowledge platform (0.768).
Collaborative affordance (CR = 0.842; AVE = 0.573; Cronbach's α = 0.842)	*In our company, GenAI is used to...* • ...implement collaboration within the organization (0.810). • ...promote information sharing between departments (0.653). • ...achieve synchronous, real-time collaborative work (0.753). • ...enable organizational members to work collaboratively (0.802).
Creational affordance (CR = 0.802; AVE = 0.574; Cronbach's α = 0.801)	*In our company, GenAI is used to...* • ...automatically produce various types of content, including text, images, and videos (0.809). • ...integrate with existing content management systems and workflows (0.732). • ...summarize documents and draft client materials (0.730).
Competitive intensity (CR = 0.887; AVE = 0.611; Cronbach's α = 0.884)	• Competition in our industry is cutthroat (0.736). • There are many "promotion wars" in our industry (0.759). • Anything that one competitor can offer, others can match readily (0.743). • Price competition is a hallmark of our industry (0.885). • One hears of a new competitive move almost every day (0.777).
Competitive advantages (CR = 0.819; AVE = 0.531; Cronbach's α = 0.819)	• Our company's earnings before interest and tax (EBIT) is higher than the industry average (0.757). • Our company's return on investment (ROI) is higher than the industry average (0.696). • Our company's return on sales is higher than the industry average (0.756). • Our company has a large market share (0.703).

Note: Numbers in parentheses are factor loadings

Fig. A1. Measurement items

Data availability

Data will be made available on request.

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